TAYTEN BENNETSEN

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EDUCATION

Rice University — Houston, Texas

B.S. in Computer Science and B.A. in Cognitive Science — Expected Graduation, April 2027

- **GPA:** 3.84/4.00
- **Concentrations:** Data Structures and Algorithms, Software Engineering, Machine Learning, OOP, Operating Systems

EXPERIENCE

Software Engineer Intern | Melody AI — December 2024 - Present

Building full-stack workflow solutions for music licensing and artist discovery — Next.js | Python | Firebase | OpenAI API

- Developing a comprehensive admin system and licensing workflow platform using **Next.js** and **Firebase**, streamlining the process of artist discovery and music integration for supervisors.
- Building **OpenAI API** integrations for automated email generation and intelligent artist matching algorithms, enhancing supervisor efficiency.
- Architecting scalable backend infrastructure in **Python** to support concurrent user operations and real-time data processing.
- Contributing to both foundational development and user-facing features through **weekly sprints** and deliverables in an **agile** environment.

Software Engineer Intern | Wostmann & Associates — May 2024 - Aug 2024

Created open source screen reader for visually impaired users — C++ | WinAPI | Microsoft UI Automation | Windows Speech API

- Designed **SightSpeak Reader** using C++, integrating WinAPI, Microsoft UI Automation, and Windows Speech API to enable real-time text-to-speech output and visual text highlighting, supporting over **4,000** Microsoft applications.
- Developed **8** efficient keyboard shortcuts and utilized Breadth-First Search (BFS) tree traversal for seamless navigation through UI elements, improving navigation speed by **80%** for beta users.
- Leveraged a **thread pool** and **futures/promises** to manage multiple tasks concurrently, reducing text processing latency by **28ms** and ensuring a smooth user experience.
- Implemented **Tesseract OCR** to enable text extraction from images and PDFs, achieving a **98%** success rate.

PROJECTS

Full-Stack Project | Bunkmate — September 2024 - Present

Rice University centralized student housing and subletting platform — Next.js | TypeScript | Supabase | Tailwind CSS

- Developing a comprehensive housing platform for Rice University students and alumni using **Next.js** and **TypeScript**, enabling seamless posting and discovery of sublets and off-campus housing.
- Implementing secure authentication system with role-based access control using **Supabase Auth**, ensuring data privacy and user functionality.
- Architecting real-time database operations through **Supabase** for favorites, listings management, and user profiles, creating a seamless and interactive user experience.

AI LangChain Project | Enhanced Security Assistant — October 2024 - November 2024

Vulnerability analysis and security research tool — Python | LangChain | OpenAI API | NVD API | Pydantic | React

- Developed a comprehensive security research tool integrating the **National Vulnerability Database (NVD) API** with **LangChain** and **OpenAI**, enabling natural language queries for vulnerability analysis across **4** distinct tool functions.
- Implemented robust error handling and rate limiting, achieving **99.9%** uptime during testing and reducing API failures by **95%** through intelligent request management.
- Designed a modular architecture using **Pydantic** for type safety, decreasing input-related errors by **85%** and improving code maintainability.

Full-Stack Project | Pillora – Medication Management Mobile Application — September 2024 - October 2024

Developed during HackRice hackathon — JavaScript | React | Node.js | MongoDB | Tailwind | OpenAI | OpenFDA | Express

- Led a team of **4** to build a cross-platform mobile app using **React** and **Node.js**, targeting an increase in medication adherence.
- Implemented **Retrieval-Augmented Generation (RAG)** by integrating the **OpenFDA API** with **OpenAI**, enhancing the AI-driven chatbot's accuracy by **35%** and reducing hallucinations by **90%** during internal testing.
- Architected a scalable backend using **MongoDB** and **Express**, capable of handling up to **250** concurrent users, laying the groundwork for future feature expansions and a projected user base growth of **200%**.
- Received **2nd place overall** and **1st in the healthcare path** at **HackRice 2024**, recognizing excellence in software innovation.

Rice Apps Dev | Software Engineering Club — September 2023 - Present

- Participated in weekly Sprints, refining software engineering skills through iterative development and problem-solving.
- Used JavaScript, React, CSS and HTML to create a Portfolio, Shopping List API Bunkmate and RDT Ticketing.

TECHNICAL SKILLS

Programming | C/C++, JavaScript, Java, Python, React, React Native, CSS, HTML, Python, SQL, R, Pandas
Tools | MongoDB, Supabase, Firebase, Next.js, Node.js, Express.js, Git, OpenAI, Anthropic, LangChain, Auth0, PyTorch, OCR, Postman, Docker